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ASSIGNMENT - 5

Section - A

Que 2 Define Partitioned clustering in data mining

Ans Partioned clustering is a method that divides a dataset of n objects in k partition where each partition represents a cluster. Each object belong to exactly one group.

Que 3 Define Web Content mining.

Ans Web Content mining refers to the extraction of useful information from the actual content of web pages. This includes mining text, images, audio and video data contained with website to find pattern.

Que 2 What is web structure mining ? Mention one typical application

Ans Web structure mining involves analyzing the node and connection structure of a website to understand the underlying topology of the web. It focus on it focus on how pages are linked to one another.

Section - B

Que 2 Explain the basic phases of text mining. Discuss text preprocessing.

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Ans Text mining involves converting unstructured text into structured data to derive insight

- Text Preprocessing → This step cleans the raw text. Techniques include Tokenization, Stop-word removal

- Document Representation - To be analyzed by algorithms, text must be converted into numbers. Common methods include the Bag of words

Text Mining Tasks →

(i) Text Classification → Assigning predefined categories to text documents

(ii) Sentiment Analysis → Determining the emotional tone behind the text

Ans What are grid-Based clustering methods Explain the general approach and discuss any one algorithm

Ans These method quantize the object space into finite number of cells that a form a grid structure. All clustering operations are performed on the grid

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(i) Divide the data space into a grid of rectangular cells.

(ii) Calculate statistical information for each cell

(iii) Locate high density cells to identify potential cluster centers.

Section - C

Q15 Write detailed notes on spatial and multimedia data mining.

Ans Spatial Data mining →

- Characteristics → Spatial data related to location and geometry. It is distinct

- Techniques →

(i) Spatial Clustering → methods like DBSCAN to find regions of high activity

(ii) Spatial Trend Analysis → detecting movement of patterns

- Applications →

- GIS → Land use planning
- Remote Sensing → analysing complex spatial data types and the high computational cost of spatial join

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Multimedia Data Mining →

- Characteristics : Includes unstructured audio, video, images and text. It involves huge data volumes

- Techniques →

(i) Feature Extraction → Converting images to color histograms or texture vectors.

- Applications →

(i) medical imaging : Detecting tumors in MRI / X-ray scan automatically

(ii) Surveillance → Detecting suspicious activity in CCTV footage